

Output Form (OF)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: MILU | Context: Hindi-Medium Competitive Exam Aspirants — North India (Central Exams)

Definition: Whether the expected output modality matches regional deployment needs and accessibility.

Justification

Output form is HIGH-priority per elicitation. The deployment requires open-ended Hindi explanatory text alongside a verdict label. MILU evaluates models exclusively via MCQ label accuracy: log-likelihood option selection for non-API models [Q52, Q53, Q54, Q55] and JSON-structured single-answer generation for API models [Q56, Q57]. The benchmark authors themselves explicitly flag this as a limitation: 'our evaluation primarily relies on the log-likelihood approach, which may yield different results compared to other established methods, such as generation-based evaluation and chain-of-thought prompting' [Q85]. The dataset analysis confirms this empirically — no rationale or explanation field exists in any of 245 reviewed examples [CRITICAL Concern 1]. Web research confirms no Hindi-language exam-rationale evaluation benchmark currently exists [WEB-14, WEB-13]. MILU's output form fundamentally cannot validate the deployment's core output: fluency, factual accuracy, or pedagogical quality of Hindi-language rationales.

ID	Evidence
WEB-14	2025 Hindi LLM benchmarking preprint notes Hindi instruction-following/conversational benchmarks are 'largely unavailable publicly'
WEB-13	IndicGenBench (ACL 2024) covers Hindi generation but not exam rationale quality
CRITICAL Concern 1	Every example in the sample is structured as a 4-option MCQ with a single correct label ('target'). There are no rationale fields, explanation columns, or any annotation supporting the deployment's core output requirement: a substantive Hindi-language explanation of why an answer is correct or incorrect.

Strengths

- Reproducible MCQ scoring via LM Evaluation Harness [Q52]
- 0-shot, 1-shot, 5-shot evaluation protocols for sensitivity analysis [Q50]

Checklist

Checklist item definitions:

ID	Assessment Question
OF-1	Does output modality match regional deployment needs?
OF-2	Assess text-to-speech availability for speech output
OF-3	Consider literacy rates and accessibility requirements
OF-4	Document form mismatches harming external validity

Checklist responses:

OF-1: Mismatch — deployment requires Hindi free-text explanation alongside verdict [elicitation Q3]; MILU produces only MCQ accuracy [Q55, CRITICAL Concern 1].

ID	Evidence
CRITICAL Concern 1	Every example in the sample is structured as a 4-option MCQ with a single correct label ('target'). There are no rationale fields, explanation columns, or any annotation supporting the deployment's core output requirement: a substantive Hindi-language explanation of why an answer is correct or incorrect.

OF-2: Not applicable — deployment is text-only; no TTS requirement.

OF-3: Target population is graduate-level and near-universally literate in Hindi; literacy is not the gating concern. The gating concern is Hindi NLG quality, which MILU does not evaluate.

ID	Evidence
WEB-13	IndicGenBench (ACL 2024) covers Hindi generation but not exam rationale quality

OF-4: Documented harms: MCQ-only scoring [Q55, Q85] cannot validate Hindi rationale fluency, coherence, or pedagogical fit; no Hindi exam-rationale benchmark exists [WEB-14] to supplement.

ID	Evidence
WEB-14	2025 Hindi LLM benchmarking preprint notes Hindi instruction-following/conversational benchmarks are 'largely unavailable publicly'
CRITICAL Concern 1	Every example in the sample is structured as a 4-option MCQ with a single correct label ('target'). There are no rationale fields, explanation columns, or any annotation supporting the deployment's core output requirement: a substantive Hindi-language explanation of why an answer is correct or incorrect.

Requires Expert Verification

- Design and validation of a Hindi rationale-quality rubric (factual accuracy, fluency, pedagogical clarity, register) requires Hindi-medium pedagogy experts

Remediation

Gap 1: MCQ-only scoring (log-likelihood / JSON option selection) cannot validate Hindi free-text rationale [Q55, Q85]

Recommendation: Add a generation-based Hindi rationale evaluation using LLM-as-judge with the rubric above, plus periodic human evaluation; consider adapting IFEval-Hi/MT-Bench-Hi protocols [WEB-14] to exam-specific prompts.